Epistemic Dynamics, Bayesian Convergence, and the Limit Object of Maximal Correctness

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Abstract

We formalize an integrated epistemic framework combining Bayesian inference, social attractor dynamics, layered models of reality, and a limiting object of maximal epistemic correctness G. We interpret epistemology as a dynamical system in a weighted vector space of belief states, incorporating insights from philosophy of science, statistics, decision theory, and theoretical computer science.

The paper ends with "The End"

1 Introduction

We consider epistemic agents operating in a structured environment consisting of layered constraints: physical (L1), emergent (L2), and narrative (L3). Belief formation is modeled as a hybrid process combining Bayesian updating and social attractor dynamics.

We define a limit object G as the idealized state of maximal epistemic correctness.

2 Epistemic State Space

Let each agent A_i have a belief state:

$$B_{i,t} \in \mathbb{R}^n$$

Let the ideal correctness state be:

$$G = (g_1, g_2, \ldots, g_n)$$

We define a weighted epistemic distance:

$$d(B_{i,t}, G) = \sqrt{\sum_{k=1}^{n} w_k (B_{i,t}^{(k)} - g_k)^2}$$

where w_k encodes layer importance.

3 Layer Structure

Layer	Domain	Epistemic Role
L1	Physical laws	Strong constraint / high fidelity inference
L2	Emergent systems	Medium constraint / statistical structure
L3	Narrative systems	Low constraint / high social modulation

Table 1: Epistemic layers and their roles

4 Update Dynamics

We define belief updates as a convex combination of Bayesian and social processes:

$$B_{i,t+1} = (1-\alpha)\,\mathcal{B}(B_{i,t}, E_{i,t}) + \alpha\,\mathcal{S}(B_{i,t}, S_t)$$

where:

- $\mathcal{B}$ is the Bayesian update operator,
- $\mathcal{S}$ is the social attractor operator,
- $\alpha \in [0,1]$ is the social weighting parameter.

5 Bayesian Update Rule

For hypothesis space $\mathcal{H}$ and evidence E:

$$P(h|E) = \frac{P(E|h)P(h)}{\sum_{h' \in \mathcal{H}} P(E|h')P(h')}$$

This induces a projection:

$$\mathcal{B}(B_{i,t}, E_{i,t}) \to B_{i,t}^{\text{posterior}}$$

6 Social Attractor Dynamics

We model social influence as:

$$\mathcal{S}(B_{i,t}, S_t) = B_{i,t} + \beta(S_t - B_{i,t})$$

where S_t represents a local epistemic attractor (consensus, institution, ritual structure).

7 Distance to the Limit Object

We define alignment probability:

$$P(B_{i,t} \sim G) = e^{-\lambda d(B_{i,t}, G)}$$

where $\lambda > 0$ is a sensitivity parameter.

8 Layer-Weighted Update Equation

Each coordinate evolves as:

$$B_{i,t+1}^{(k)} = B_{i,t}^{(k)} + (1-\alpha) w_L \Delta\mathcal{B}^{(k)} + \alpha w_L \Delta\mathcal{S}^{(k)}$$

where $w_L \in \{w_{L1}, w_{L2}, w_{L3}\}$.

9 Computational Interpretation

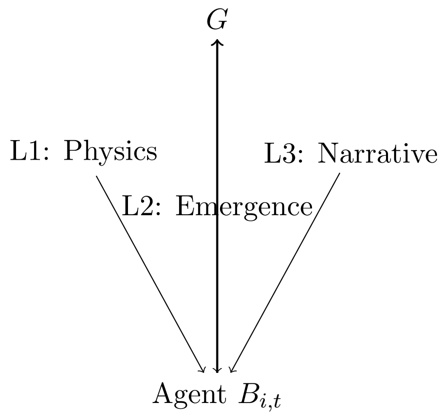
The system defines a stochastic dynamical system:

$$B_{t+1} = F(B_t, E_t, S_t)$$

with:

- Bayesian inference as likelihood maximization,
- Social dynamics as gradient toward attractors,
- Physical constraints as external conditioning.

10 Simulation Diagram



11 Algorithmic Formulation

Algorithm 1: Epistemic Update

1. Initialize belief vector $B_{i,0}$
2. Observe evidence $E_{i,t}$
3. Compute Bayesian update $\mathcal{B}$
4. Compute social influence $\mathcal{S}$
5. Update:
$$B_{i,t+1} = (1 - \alpha)\mathcal{B} + \alpha\mathcal{S}$$
6. Repeat

12 Interpretation via AI and Data Science

This system is equivalent to:

- Bayesian filtering in probabilistic graphical models
- Gradient descent on a composite loss function:
$$\mathcal{L} = d(B, G) + \alpha \cdot \mathcal{R}_{social}$$
- Multi-agent reinforcement learning with shared latent structure

13 Philosophical Interpretation

We interpret G as a regulative limit object:

- In philosophy: ideal truth condition
- In mathematics: consistency closure of all provable systems
- In theology: maximal epistemic divinity
- In statistics: zero-error posterior limit

14 Glossary

G Limit object of maximal epistemic correctness

$B_{i,t}$ Belief state of agent i at time t

$L1$ Physical constraint layer

$L2$ Emergent system layer

$L3$ Narrative/social layer

α Social influence weight

λ Sensitivity of correctness alignment

$\mathcal{B}$ Bayesian update operator

$\mathcal{S}$ Social attractor operator

15 Conclusion

Epistemic agents evolve in a layered constraint system where Bayesian inference and social dynamics jointly define trajectories in belief space. These trajectories can be interpreted as asymptotic approximations to a limit object G, representing maximal epistemic correctness across all representable domains.

References

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The End